

16. (2 points)

Given the following information (\$000) as of December 31, 2016:

Accident <u>Year</u>	Earned Risk Pure <u>Premium</u>	Rate-Level Adjusted <u>Premium</u>	Aggregate Reported <u>Loss</u>	Aggregate Loss <u>Report Lag</u>	Chain Ladder <u>IBNR</u>	Bornhuetter- Ferguson <u>IBNR</u>
2012	5,000	5,000	3,500	.91	350	364
2013	4,500	5,500	3,300	.80	825	900
2014	5,000	6,000	2,750	.67	1,375	1,667
2015	6,500	6,500	2,500	.40	3,750	3,300
2016	7,000	7,000	2,000	.20	8,000	4,800

a. (1 point)

Calculate IBNR for accident year 2016 using the Stanard-Bühlmann method.

b. (0.5 point)

Briefly describe one advantage and one disadvantage of the Stanard-Bühlmann method.

c. (0.5 point)

Comment on an appropriate IBNR selection for accident year 2016.

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QUESTION 16

TOTAL POINT VALUE: 2

LEARNING OBJECTIVE: A11, A12, A13

SAMPLE ANSWERS

Part a: 1 point

Sample 1

Acc. Year	Adj. EP	Report Lag	Used up EP	Reported Loss
2012	5,000	.91	4,550	3,500
2013	5,500	.80	4,400	3,300
2014	6,000	.67	4,620	2,750
2015	6,500	.40	2,600	2,500
2016	7,000	.20	1,400	2,000
Total			16,970	14,050

$$\text{ELR} = 14,050/16,970 = .828$$

$$\text{AY 2016 IBNR} = \text{Adj. EP} * \text{ELR} * (1 - \text{Report Lag}) = 7000 * .828 * .8 = 4637$$

Sample 2

$$\text{Used-up EP} = (5000 * .91 + 5500 * .80 + 6000 * .67 + 6500 * .4 + 7000 * .2) = 16,970$$

$$\text{Total reported loss} = (3500 + 3300 + 2750 + 2500 + 2000) = 14,050$$

$$\text{ELR} = 14050/16970 = .828$$

$$\text{IBNR for AY 2016} = 7000 * .828 * (1 - .20) = 4637$$

Part b: 0.5 point

Sample 1

- Advantage: more stable than CL at early maturities where CDF would be highly leveraged and losses volatile
- Disadvantage: need to adjust premiums for rate changes and it can be difficult to obtain accurate on-level EP.

Sample 2

- Advantage: SB uses actual experience, including latest year, to determine the ELR.
- Disadvantage: The rate-level adjustment can be difficult, depending on information available.

Sample 3

- Advantage: calculates the ELR from the data rather than using a selection like the BF method.
- Disadvantage: requires rate level adjusted premium which may be difficult to obtain.

Part c: 0.5 point

Sample 1

I would use a credibility weighted IBNR since CL is highly leveraged but SB and BF are low and don't take the large reported losses for the year into account in their IBNR calculation. It could reflect a change in the level of loss for the year.

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Sample 2

AY 2016 has reported losses that are much higher than expected based on older years. We would have to give that some credibility, so I would use some type of credibility-weighted method between the SB and chain ladder. I would give little credibility to the reported losses, especially if it was due to one large claim.

Sample 3

AY	Chain-ladder LR
2012	.77
2013	.75
2014	.69
2015	.96
2016	1.43

For 2017, losses at 2m seem very high looking at chain ladder results. Given that only 20% is assumed reported, it might be better not to rely on the experience using chain ladder. I prefer the Bornhuetter-Ferguson here because this seems like an uncertain year. Since the BF is higher than SB, it is a more conservative choice. The BF a priori ELR might reflect some additional information about this book of business that is not reflected in the SB method.

EXAMINER'S REPORT

Candidates were expected to know the definition, assumptions, calculation procedure and relative advantages for the chain ladder, Bornhuetter-Ferguson and Stanard-Bühlmann reserving methods. Candidates were also expected to be able to apply this knowledge to a given set of data.

Part a

Candidates were expected to know how to calculate the Stanard-Bühlmann method, including the use of adjusted earned premium to calculate the used-up premium, calculation of the expected loss ratio (ELR) and calculation of IBNR.

Common errors include:

- Confusing the SB method with a credibility method.
- Deriving the IBNR by taking the SB indicated ultimate with full adjusted premium and subtracted reported losses to date rather than using the unused premium portion to derive the IBNR.

Part b

Candidates were expected to know the advantages and disadvantages of the Stanard-Bühlmann method.

Common errors include:

- Listing the same issue as an advantage and disadvantage. For example, stating that data was used to compute the ELR as an advantage and stating that the method was not responsive to changes in the data as a disadvantage.
- Commenting that two actuaries would come to the same result using the same data. This is not actually true in all cases. However, if the candidate responded with two actuaries coming up with the same result given the same adjusted premium, then this was a valid

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response.
Part c
Candidates were expected to know how to select and justify the choice of a method for establishing the IBNR reserve based on a given set of loss data. Full credit was given for responses that commented both on the choice of method and how some aspect of the data supported this choice. Common errors include: • Stating the relative advantages of the methods without explaining how their recommendations fit the data. • Recommending taking the middle estimate or the average of the two low estimates without justification by some aspect of the data.